# Goldbach-type sums $a + b = N$ with $\Omega(ab) \leq 32$

Giovane de Morais

*Centro de Competência em Manufatura (CCM), Graduate Program in Aeronautical and Mechanical Engineering, Instituto Tecnológico de Aeronáutica (ITA), São José dos Campos, SP, Brazil*

**Abstract**

We prove that every integer $N \geq 2$ can be written as $N = a + b$ with positive integers $a, b$ such that $\Omega(ab) \leq 32$, where $\Omega$ counts prime factors with multiplicity. This improves the bound $\Omega(ab) \leq 33$ of Campbell. The argument stays inside Campbell's framework, the explicit Friedlander–Iwaniec lower-bound sieve applied to the products $n(N - n)$. Three elementary reductions show that only odd $N$ need a new estimate, with sifting level $z = (N^2/8)^{1/32}$. We close that case with a per-interval dimension constant, per-interval sieve parameters and pre-sieving by both 3 and 5. The intermediate range is certified in ball arithmetic over 117 415 intervals, and the large range by a monotonicity argument. The proof uses the dimension condition at the sifting level, the same reading on which Campbell's bound relies.

*Keywords:* Almost primes, Goldbach-type theorems, sieve methods, explicit bounds, ball arithmetic
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## 1. Introduction

Let $\Omega(n)$ denote the number of prime factors of $n$ counted with multiplicity. Chen's theorem [1] represents every sufficiently large even integer as a prime plus an integer with at most two prime factors, and Rényi's theorem [2] gives a uniform bound for all even $N \geq 4$ with an unspecified constant. Johnston and Starichkova [3] made Rényi's theorem explicit: every even $N \geq 4$ is $p + r$ with $p$ prime and $\Omega(r) \leq 395$, and $\Omega(r) \leq 31$ under the generalised Riemann hypothesis. Their proof uses the explicit linear sieve of Bordignon, Johnston and Starichkova [4].

Relaxing the condition that one summand be prime, one asks for $N = a + b$ with $\Omega(ab)$ small. Dudek and Dunn [5] proved $\Omega(ab) \leq 40$ for every $N \geq 2$, applying the explicit Friedlander–Iwaniec $\Lambda^-\Lambda^2$ lower-bound sieve [6]. In their setting the extremal case $w = 3$ of the dimension condition forces the constant $K = 3$. Campbell [7] removed this case by pre-sieving at the prime 3 and obtained $\Omega(ab) \leq 33$. He also wrote that $\Omega(ab) \leq 32$ "appears to be within reach" and listed four possible refinements: $K$ computed per interval; $s$ and $\delta$ optimised per interval; pre-sieving by 3 and 5; sharper constants [7, §6].

**Theorem 1.1.** *Assume Hypothesis 4.1, i.e. Theorem 2.2 with the dimension condition* (2.1) *required at the sifting level under consideration. Then every integer $N \geq 2$ can be written as $N = a + b$ with $a, b$ positive integers and*

$$\Omega(ab) \leq 32.$$

Hypothesis 4.1 is the reading of the dimension condition stated in [7, (2.1)] ("for each value of $z$ under consideration"), and Campbell's bound $\Omega(ab) \leq 33$ already depends on it: the global reading, over all pairs $2 \leq w < z'$, forces $K \geq 2$, which is incompatible with the constant $K = 1.146$ of [7, Lemma 4.1] (Remark 4.2). Thus Theorem 1.1 requires no hypothesis beyond the one used in [7].

*Email address:* giovanedms@gmail.com (Giovane de Morais)

*Where* 33 *comes from.* The bound 33 comes only from odd $N$. With $z = N^{1/r}$ and $a_n < N^2$, [7, Lemma 5.2] gives $\Omega(ab) < 2r$ for even $N$ and $\Omega(ab) < 2r + 1$ for odd $N$; with $r = 16.5$ this is $\le 32$ and $\le 33$ respectively. Three elementary observations (Section 3) reduce the problem to odd $N$: for even $N$ beyond the prime-gap range, Campbell's proof already gives $\Omega(ab) \le 32$; the prime-gap lemma [7, Lemma 5.1] holds under $1854m < 2^{M-1}$ instead of $1854m \le 2^{M-2}$, which covers $N \le N_0$ with $M = 32$; and for odd $N$ one has $a_n \le (N^2 - 1)/8$, so the level $z = (N^2/8)^{1/32}$, i.e. $N = \sqrt{8}\, z^{16}$, suffices. The odd case at this level is essentially Campbell's "$r = 16$" with an extra factor $\sqrt{8}$ in the main term. With his uniform constants, Campbell estimated that the analytic range would then start only at $N$ of order $10^{192}$, with a finite range up to $z$ of order $10^{12}$ [7, §6]. We close it with the first three refinements listed above; no new sieve theory is needed. The per-interval constant is essential (Section 6.2).

*Organisation.* Section 2 recalls Campbell's framework, Section 3 proves the reductions, Section 4 gives the sieve input and the pre-sieving by $\{3, 5\}$, and Section 5 proves Theorem 1.1. Section 6 explains why the method does not give $\Omega(ab) \le 31$.

## 2. Notation and Campbell's framework

Throughout, $p$ and $q$ denote primes, $\omega(n)$ is the number of distinct prime factors of $n$, and $\tau_4(n) = \#\{(d_1, d_2, d_3, d_4) \in \mathbb{Z}_{>0}^4 : d_1 d_2 d_3 d_4 = n\}$. We follow [7, §§2–3].

### 2.1. The sieve

Let $\mathcal{A} = (a_n)_{n \in I}$ be a finite indexed sequence of positive integers and $\mathcal{P}$ a set of sifting primes. For square-free $d$ composed of primes of $\mathcal{P}$ put $\mathcal{A}_d = (a_n)_{n \in I,\, d \mid a_n}$ and suppose

$$|\mathcal{A}_d| = X g(d) + r_d,$$

where $X > 0$ and $g$ is multiplicative with $0 < g(p) < 1$ for $p \in \mathcal{P}$. For $z \ge 2$ let $P(z) = \prod_{p<z,\, p \in \mathcal{P}} p$, $S(\mathcal{A}, z) = \#\{n \in I : (a_n, P(z)) = 1\}$ and $V(z) = \prod_{p \mid P(z)} (1 - g(p))$.

**Definition 2.1** ([7, (2.1)])**.** Let $\kappa \ge 1$ and $K \ge 1$. The function $g$ satisfies the dimension condition with parameters $\kappa$ and $K$ if, for each value of $z$ under consideration and every $2 \le w < z$,

$$\prod_{\substack{w \le p < z \\ p \in \mathcal{P}}} (1 - g(p))^{-1} \le K \Big( \frac{\log z}{\log w} \Big)^{\kappa}. \tag{2.1}$$

**Theorem 2.2** ([7, Theorem 2.1]; from [6, Theorems 7.7 and 7.1])**.** *With the notation above, let $D \ge z$, $s = \log D / \log z$ and $k = \kappa + \log K$. If $s \ge 2k + 3$, then*

$$S(\mathcal{A}, z) \ge X V(z) F(s, k) - 2 R_4(\mathcal{A}, D),$$

*where*

$$F(s, k) = 1 - \frac{s+3}{2e^k} \Big( \frac{2ek}{s-3} \Big)^{(s-3)/2}, \qquad R_4(\mathcal{A}, D) = \sum_{\substack{d \mid P(z) \\ d < D}} \tau_4(d)\, |r_d|. \tag{2.2}$$

We use Theorem 2.2 as stated in [7].

*2.2. The sequence*

Fix $N \geq 2$. For $1 \leq n \leq N-1$ let $a_n = n(N-n)$ if $N$ is even and $a_n = n(N-n)/2$ if $N$ is odd, let $\mathcal{B} = (a_n)_{1 \leq n \leq N-1}$ and $Y := N-1$ [7, §3]. The local density is the multiplicative function with

$$g(2) = \tfrac{1}{2}, \qquad g(p) = \tfrac{2}{p} \ (p \nmid N), \qquad g(p) = \tfrac{1}{p} \ (p \mid N), \qquad p \text{ odd},$$

and for square-free $\ell$ the remainder $\rho_\ell$ of the auxiliary sequence is defined by

$$|\mathcal{B}_\ell| = Y g(\ell) + \rho_\ell, \qquad \rho_1 = 0, \tag{2.3}$$

which is [7, (3.1)]. We write $g^*(2) = \frac{1}{2}$ and $g^*(p) = 2/p$ for odd $p$; then $g \leq g^*$ pointwise, uniformly in $N$.

**Lemma 2.3** ([7, proof of Lemma 4.3], via [8, Theorem 7.2])**.** *For every square-free $\ell$, $|\rho_\ell| \leq 2^{\omega(\ell)}$.*

*2.3. Pre-sieving*

Let $Q = \{3\}$ (Campbell) or $Q = \{3, 5\}$ (this paper). Put

$$I_Q = \{1 \leq n \leq N-1 : \ q \nmid a_n \text{ for all } q \in Q\}, \qquad \mathcal{A} = (a_n)_{n \in I_Q},$$

$X = |I_Q|$ and $\mathcal{P} = \{p : \ p \notin Q\}$. For $Q = \{3\}$, Campbell shows $|\mathcal{A}_d| = Xg(d) + r_d$ with $r_d = \rho_d - \rho_{3d} + g(d)\rho_3$ [7, (3.3)], the bound $|r_d| \leq \frac{19}{6} 2^{\omega(d)}$ and $X \geq (N-1)/3$. The case $Q = \{3, 5\}$ is treated in Section 4.2. In both cases we write $C_Q$ for a constant with $|r_d| \leq C_Q 2^{\omega(d)}$ for $d \mid P(z)$, $d > 1$; thus $C_{\{3\}} = 19/6$ and $C_{\{3,5\}} = 11$.

Since $d \mid P(z)$ is square-free, $\tau_4(d) 2^{\omega(d)} = \tau_8(d)$, and Rankin's trick gives, for $0 < \delta < 1$ [7, (4.4)],

$$R_4(\mathcal{A}, D) \leq C_Q \, D^\delta \prod_{\substack{p < z \\ p \in \mathcal{P}}} \Big(1 + \frac{8}{p^\delta}\Big). \tag{2.4}$$

*2.4. Campbell's proof*

Campbell sets $N_0 = 11746518036554764849893443 9$, $z = N^{1/16.5}$, $z_0 = N_0^{1/16.5} = 38.018\ldots$, $z_1 = 10^8$ and $N_1 = 10^{132}$, and splits the proof into three ranges [7, §§4–5]: Case 1, $2 \leq N \leq N_0$, by prime-gap data ([7, Lemma 5.1]); Case 2, $z_0 < z < z_1$, by a floating-point finite verification with $K = 1.146$, $s = 14.8$ and $\delta \in \{0.20, 0.60, 0.75, 0.85, 0.93\}$; Case 3, $N \geq N_1$, analytically with $K = 1.097$, $s = 14.66$ and $F > 0.0563$. Cases 2 and 3 establish $S(\mathcal{A}, z) > 0$, and [7, Lemma 5.2] converts this into the bound on $\Omega(ab)$.

*2.5. Inputs taken from the literature*

Besides Theorem 2.2 and Lemma 2.3, the proof uses:

(H3) Rosser–Schoenfeld [9, Theorem 5]: $\sum_{p \leq x} 1/p > \log\log x + B - \frac{1}{2\log^2 x}$ for $x > 1$, and $\sum_{p \leq x} 1/p < \log\log x + B + \frac{1}{2\log^2 x}$ for $x \geq 286$.

(H4) $\sum_{p \leq x} \frac{\log p}{p} < \log x$, as in [7, §4] ("corollary to Theorem 6" of [9]).

(H5) The Meissel–Mertens constant satisfies $B = 0.26149721284764\ldots < 0.2614972128477$.

(H7) Every prime $q < G := 101412319996363109 23$ is followed by a prime $q'$ with $q' - q \leq 1854$ [10], as used in [7, §5].

## 3. Reductions

### *3.1. The prime-gap range*

**Lemma 3.1.** *Let $m$ be 1 or a prime and $M$ a positive integer with $1854m < 2^{M-1}$. Then for every $2 \le N \le Gm$ there are positive integers $a, b$ with $N = a + b$ and $\Omega(ab) \le M$.*

*Proof.* Every prime factor is at least 2, so $2^{\Omega(x)} \le x$ for every integer $x \ge 1$. Hence if $1 \le x \le 1854m < 2^{M-1}$, then $\Omega(x) \le \log_2 x < M - 1$, and since $\Omega(x)$ is an integer, $\Omega(x) \le M - 2$.

If $N \le 2m$, take $a = 1$, $b = N - 1 \ge 1$. Then $b < 2m \le 1854m$, so $\Omega(ab) = \Omega(N - 1) \le M - 2 < M$.

If $N > 2m$, let $p$ be the largest prime with $p < N/m$ (it exists since $N/m > 2$) and $p'$ the next prime, so that $p < N/m \le p'$. Since $p < N/m \le G$, (H7) gives $p' - p \le 1854$. Hence $0 < N - mp \le m(p' - p) \le 1854m$ and $\Omega(N - mp) \le M - 2$. With $a = mp$ and $b = N - mp$ we have $\Omega(mp) \le 2$, so $\Omega(ab) \le M$. □

**Corollary 3.2.** *For every $2 \le N \le N_0$ there are $a, b \ge 1$ with $N = a + b$ and $\Omega(ab) \le 32$.*

*Proof.* Take $M = 32$ and the prime $m = 1158293$. Then

$$1854m = 2147475222 < 2^{31} = 2147483648, \qquad Gm = N_0,$$

and Lemma 3.1 applies. □

*Remark* 3.3. [7, Case 1] writes $1854m = 2147475222 < 2^{31} = 2^{M-2}$ with $M = 33$. The gain is only the rounding of $\log_2$ (the strict inequality for $N \le 2m$ is already in [7]). The slack is necessary: $1854m \le 2^{30} = 1073741824$ is false, so the hypothesis of [7, Lemma 5.1] cannot be used with $M = 32$ and this $m$.

### *3.2. Even N*

**Proposition 3.4.** *For every even $N > N_0$ there are $a, b \ge 1$ with $N = a + b$ and $\Omega(ab) \le 32$.*

*Proof.* Cases 2 and 3 of the proof of [7, Theorem 1.2] establish $S(\mathcal{A}, z) > 0$ with $Q = \{3\}$ and $z = N^{1/16.5}$ for every $N > N_0$. No step uses the parity of $N$: the bound $X \ge (N - 1)/3$ holds for all $N$, and the bounds $g(2) = \frac{1}{2}$, $g(p) \le 2/p$ do not depend on $N$. [7, Lemma 5.2] with $r = 16.5$ then gives $\Omega(ab) < 33$ for even $N$; Campbell states this explicitly in Cases 2 and 3. □

*Remark* 3.5. Proposition 3.4 inherits Campbell's finite verification, which is carried out in floating point with a cushion of 0.10 in the logarithm. Reproducing the three scripts of [11] (tag `arxiv-v1`) gives output identical to the versioned output, with 227 intervals and minimal log-ratio 0.1000056. We have not certified that computation.

### *3.3. Odd N*

**Lemma 3.6.** *If $N$ is odd and $1 \le n \le N - 1$, then $a_n = n(N - n)/2 \le (N^2 - 1)/8 < N^2/8$.*

*Proof.* The function $n \mapsto n(N - n)$ is concave and symmetric about $N/2$. Over the integers its maximum is at $n = (N \pm 1)/2$, where it equals $\frac{N-1}{2} \cdot \frac{N+1}{2} = \frac{N^2-1}{4}$. □

**Lemma 3.7.** *Let $Q \in \{\{3\}, \{3, 5\}\}$, let $N > N_0$ be odd and let $z := (N^2/8)^{1/32}$, i.e. $N = \sqrt{8}\, z^{16}$. If $S(\mathcal{A}, z) > 0$, then there are $a, b \ge 1$ with $N = a + b$ and $\Omega(ab) \le 32$.*

*Proof.* Since $N > N_0$ we have $z > z_0^* := (N_0^2/8)^{1/32} = 39.91606854\ldots$; in particular $z > 5$. As $S(\mathcal{A}, z) > 0$ there is $n \in I_Q$ with $(a_n, P(z)) = 1$. Since no $q \in Q$ divides $a_n$ and $\mathcal{P}$ contains every other prime, $a_n$ has no prime factor less than $z$. Let $t = \Omega(a_n)$. By Lemma 3.6, $z^t \le a_n < N^2/8 = z^{32}$, so $t \le 31$. With $a = n$, $b = N - n$ we have $ab = 2a_n$ and $\Omega(ab) = t + 1 \le 32$. □

The level $z$ depends on $N$, and no lemma of [7] uses a fixed exponent $r$ except [7, Lemma 5.2]. The bound $X \geq (N-1)/3$ continues to hold, and $z_0^* > z_0$, so the ranges of [7, Lemma 4.1] remain covered.

In the computation we start from the floating-point number

$$u_0 = 39.91606854834768,$$

for which $8u_0^{32} \leq N_0^2$ is checked in exact rational arithmetic; hence every $N > N_0$ has $z > u_0$.

**Corollary 3.8.** *Theorem 1.1 follows from Corollary 3.2, Proposition 3.4 and the statement*

$$S(\mathcal{A}, z) > 0 \quad \textit{for every odd } N \textit{ with } z = (N^2/8)^{1/32} > u_0.$$

We prove it with $Q = \{3, 5\}$. All bounds depend on $N$ only through $z$, so the proof is uniform in odd $N$.

## 4. The sieve input

### *4.1. The dimension condition at the sifting level*

**Hypothesis 4.1** (local reading)**.** Fix $N$, hence $\mathcal{A}$ and $g$, and a level $z = z(N)$. Theorem 2.2 holds at this $z$ with constants $\kappa = 2$ and $K$ provided

$$\prod_{\substack{w \leq p < z \\ p \in \mathcal{P}}} (1 - g(p))^{-1} \leq K\Big(\frac{\log z}{\log w}\Big)^2 \qquad \text{for every } 2 \leq w < z,$$

at this $z$ (not for all $z' \leq z$).

This is Definition 2.1 as written in [7, (2.1)]. Campbell's Lemma 4.1(1) is stated "for every $z \geq z_0$ and every $2 \leq w < z$", and he explicitly uses "the restrictions $z \geq z_0 \ldots$ rather than estimates uniform over a larger range" [7, §4].

*Remark* 4.2 (the global reading). Suppose (2.1) were required for all pairs $2 \leq w < z'$, for every $z'$. With $w = 2$ and $z' \to 2^+$ the product equals $(1 - g(2))^{-1} = 2$, while $(\log z'/\log 2)^2 \to 1$; this forces $K \geq 2 > 1.146$. (With $w = 5$ and $z' \to 5^+$ one obtains $K \geq 5/3$.) Hence [7] already relies on a reading that is not global. A hypothetical reading "for all $z' \in [z_0, z]$" would save [7] but not the present argument, since it amounts to a uniform constant, which fails here (Section 6.2); that reading is not natural in [6], which does not know $z_0$.

*Remark* 4.3 (evidence for the local reading). 1. In Blight's thesis [12, Definition 2.1.1, p. 6] (advisor H. Iwaniec), $g$ is supported on $P(z)$ and the condition is assumed "for any $w \geq 2$ with $w < z$", with $z$ the sifting level.

2. In [12, §2.3, pp. 13–14], with $I(X, z) = \sum_{d \mid P(z),\, d \geq X} h(d)$, $h(p) = g(p)/(1 - g(p))$ and $Y_B := \sqrt{D/z}$ (so that $s = 2\log Y_B/\log z + 1$), one has

$$J^2 V(D, z)V(z) \geq 1 - \frac{\log(Y_B z^2)}{\log z}\, I(Y_B/z, z)V(z),$$

and, citing [6, p. 111], $I(X, z)V(z) \leq e^{-\kappa}(2e\kappa/t)^{t/2}$ for $t = 2\log X/\log z > 2\kappa$. With $X = Y_B/z$ one gets $t = s - 3$ and $\log(Y_B z^2)/\log z = (s+3)/2$, which gives $1 - \frac{s+3}{2e^\kappa}\big(\frac{2e\kappa}{s-3}\big)^{(s-3)/2}$: this is $F(s, k)$ of (2.2) with $\kappa$ in place of $k$. The combinatorial step uses only $h(p) \geq g(p)$; the dimension condition enters only through the bound for $I(X, z)$, at the same $z$.

3. Friedlander [13, Appendix] writes that "the density function $g$ satisfies the condition [. . . , (5.38)] of dimension $\kappa = 2$", which identifies the condition in [6] as its equation (5.38).

4. Against: Dudek and Dunn write the condition "for all $z > w \geq 2$" and remark that a better $K$ might be obtained by reworking the proof so that a different range of $w$ and $z$ could be taken (preprint versions 1 and 4 of [5]). This suggests that the book's statement is global and that restricting it requires re-reading the proof.

Proposition 4.4 redoes the Rankin bound for $I(X,z)$ with general $K$, using the dimension condition only at the level $z$. It is supporting evidence, not used in the proof of Theorem 1.1.

**Proposition 4.4.** *Fix $z$ and let $g$ satisfy $0 < g(p) < 1$ for $p \in \mathcal{P}$. Put $\mathcal{L}_z(t) = \sum_{t \le p < z,\, p \in \mathcal{P}} -\log(1-g(p))$ and suppose*

$$\mathcal{L}_z(t) \le \kappa \log \frac{\log z}{\log t} + \log K \qquad (2 \le t < z).$$

*Let $k = \kappa + \log K$ and $u = \log X / \log z$. If $u \ge k$, then*

$$I(X,z)V(z) \le e^{-k}\Big(\frac{ek}{u}\Big)^u.$$

*With $u = t/2$ this is $e^{-k}(2ek/t)^{t/2}$; with $u = (s-3)/2$ it gives $F(s,k)$, including the condition $s \ge 2k+3$.*

*Proof.* For $\varepsilon > 0$, $I(X,z) \le X^{-\varepsilon} \prod_{p<z}(1+h(p)p^{\varepsilon})$ and $V(z)^{-1} = \prod_{p<z}(1+h(p))$. Since $(1+hp^{\varepsilon})/(1+h) = 1 + g(p)(p^{\varepsilon}-1)$,

$$I(X,z)V(z) \le X^{-\varepsilon} \prod_{p<z} \big(1 + g(p)(p^{\varepsilon}-1)\big) \le \exp\Big(-\varepsilon \log X + \sum_{p<z} g(p)(p^{\varepsilon}-1)\Big).$$

Take $\varepsilon = c/\log z$ and $f(t) = t^{\varepsilon} - 1$, which is non-negative and increasing. Since $g \le -\log(1-g)$, partial summation gives $\sum_{p<z} g(p)f(p) \le f(2)\mathcal{L}_z(2) + \int_2^z f'(t)\mathcal{L}_z(t)\,dt$. Insert the hypothesis. The $\log K$ terms give $\log K\,(f(2) + \int_2^z f') = \log K\,(e^c - 1)$. With $\varphi(t) = \log(\log z/\log t)$, $\varphi(z) = 0$, integration by parts turns the $\kappa$ terms into

$$\kappa \int_2^z \frac{t^{\varepsilon}-1}{t \log t}\,dt = \kappa \int_{\varepsilon \log 2}^{c} \frac{e^v - 1}{v}\,dv \le \kappa(e^c - 1),$$

because $(e^v-1)/v \le e^v$. The total is at most $k(e^c-1)$, so $I(X,z)V(z) \le \exp(-cu + k(e^c-1))$. Choosing $e^c = u/k$ (which needs $u \ge k$) gives the claim. □

*4.2. Pre-sieving by $Q = \{3,5\}$*

From now on $Q = \{3,5\}$, $I_{15} := I_Q$ and $\mathcal{P} = \{p : p \notin \{3,5\}\}$.

**Lemma 4.5.** *For square-free $d$ composed of primes of $\mathcal{P}$,*

$$|\mathcal{A}_d| = Xg(d) + r_d, \qquad r_d = \sum_{e \mid 15} \mu(e)\big(\rho_{de} - g(d)\rho_e\big), \qquad r_1 = 0.$$

*For $Q = \{3\}$ the same computation gives $r_d = \rho_d - \rho_{3d} + g(d)\rho_3$, which is [7, (3.3)].*

*Proof.* Since $(d,15) = 1$, $de \mid a_n$ if and only if $d \mid a_n$ and $e \mid a_n$, so inclusion–exclusion gives $|\mathcal{A}_d| = \sum_{e \mid 15} \mu(e)|\mathcal{B}_{de}|$. By (2.3) and multiplicativity, $\sum_e \mu(e)|\mathcal{B}_{de}| = Yg(d)(1-g(3))(1-g(5)) + \sum_e \mu(e)\rho_{de}$. Likewise $X = \sum_e \mu(e)|\mathcal{B}_e| = Y(1-g(3))(1-g(5)) + \sum_e \mu(e)\rho_e$. Eliminating $Y(1-g(3))(1-g(5))$ gives the identity. □

**Lemma 4.6.** *For $d \mid P(z)$ with $d > 1$, $|r_d| \le 11 \cdot 2^{\omega(d)}$.*

*Proof.* By Lemma 2.3, $|\rho_3|, |\rho_5| \le 2$ and $|\rho_{15}| \le 4$. For $d > 1$ composed of primes of $\mathcal{P}$ we have $g(d) \le \frac{1}{2}$ and $2^{\omega(d)} \ge 2$. Hence, using $\rho_1 = 0$,

$$|r_d| \le \sum_{e \mid 15} 2^{\omega(de)} + \tfrac{1}{2}\big(|\rho_3| + |\rho_5| + |\rho_{15}|\big) \le 9 \cdot 2^{\omega(d)} + 4 \le 11 \cdot 2^{\omega(d)},$$

where $9 = 1 + 2 + 2 + 4$. □

*Remark* 4.7. The analogue in [7] is $3 \cdot 2^{\omega} + \frac{1}{3} \le \frac{19}{6} 2^{\omega}$. Since $\max(|\rho_3| + |\rho_5| + |\rho_{15}|) = 10/3$ (periodic modulo 60), one could use $(9 + \frac{5}{6})2^{\omega(d)}$, i.e. $C = 9.83$, which would add about $\log(11/9.83) \approx 0.11$ to the margins. This finer value is not needed: the margins are positive with $C_Q = 11$, which is the value used throughout and in the certificate.

**Lemma 4.8.** $X \ge (N-1)/5 - 3 = (N-16)/5$.

*Proof.* By the Chinese remainder theorem, the admissible classes modulo 15 are products of admissible classes modulo 3 and modulo 5. Modulo 3, $3 \mid a_n$ if and only if $n \equiv 0$ or $n \equiv N$, so there is 1 admissible class if $3 \nmid N$ and 2 if $3 \mid N$; modulo 5 there are 3 or 4. Hence there are at least 3 admissible classes modulo 15, each containing at least $\lfloor (N-1)/15 \rfloor > (N-1)/15 - 1$ elements of $[1, N-1]$. Therefore $X > 3\big((N-1)/15 - 1\big) = (N-1)/5 - 3$. □

Theorem 2.2 applies without change, since it is stated for an arbitrary sequence and set of sifting primes, and $0 < g(p) < 1$ on $\mathcal{P}$. Moreover,

$$V(z) \ge \tfrac{1}{2} \prod_{7 \le p < z} \Big(1 - \frac{2}{p}\Big), \tag{4.1}$$

the factor $1 - g(5)$ having moved from $V$ into $X$, and (2.4) holds with $C_Q = 11$. In Lemma 3.7 we used $z > 5$.

*Remark* 4.9. Removing 5 from $\mathcal{P}$ removes the worst case of [7, Lemma 4.1] ($z \to 43^+$, $w = 5$, constant 1.1457). The new worst case is $z \to 43^+$, $w = 11$, with $K = 1.089970$ on the cell $(42.9989, 43.0032]$. Campbell anticipated this: "the prime 5, which is responsible for the largest values of $K$" [7, §6].

*4.3. The dimension constant on an interval*

**Lemma 4.10.** *Let $I = (u, v]$ with $u \ge u_0$ and let*

$$K_I \;\ge\; \sup_{z \in I} \sup_{2 \le w < z} \prod_{\substack{w \le p < z \\ p \in \mathcal{P}}} (1 - g^*(p))^{-1} \Big(\frac{\log w}{\log z}\Big)^2 .$$

*Then for every $N$ and every $z \in I$, the condition of Hypothesis 4.1 holds at $z$ with $K = K_I$, and $K_I \ge 1$. Moreover, the supremum is approached as $z \to u^+$ or $z \to q^+$ with $q \in I \cap \mathcal{P}$, and it suffices to take $w = 2$ or $w \in \mathcal{P}$.*

*Proof.* The product increases with $g$, and $g \le g^*$, so the bound is uniform in $N$. Letting $w \to z^-$ shows $K_I \ge 1$. For a fixed set of included primes the quantity decreases in $z$, so on each interval between consecutive primes of $\mathcal{P}$ its largest value is approached at the left end; as a function of $w$ the product is constant between consecutive primes while $(\log w)^2$ increases. This is the argument of Case 3 of [7, Lemma 4.1]. □

Since $K_I$ satisfies the condition at every $z \in I$, Theorem 2.2 applies at each $z \in I$ with $k_I = 2 + \log K_I$; no monotonicity of $F$ in $k$ is needed.

For the analytic range let $Z$ be an even integer, put

$$T := \frac{1}{Z-2} + \frac{1}{Z-1} \;\ge\; \sum_{n \ge Z} \frac{2}{n(n-2)}$$

(telescoping: $\frac{2}{n(n-2)} = \frac{1}{n-2} - \frac{1}{n}$), and

$$K_{\mathrm{ex}}(Z) = \sup_{2 \le w < Z} \prod_{\substack{w \le p < Z \\ p \in \mathcal{P}}} (1 - g^*(p))^{-1} \Big(\frac{\log w}{\log Z}\Big)^2 .$$

**Lemma 4.11.** *For every $z \ge Z \ge 286$ and every $2 \le w < z$, the dimension condition holds at $z$ with $\kappa = 2$ and*

$$K_{\mathrm{an}}(Z) = \max\bigl(1, K_{\mathrm{ex}}(Z)\bigr) \exp\Bigl(\frac{2}{\log^2 Z} + T\Bigr).$$

*Proof.* (i) $w \ge Z$. For odd $p \in \mathcal{P}$, $-\log(1-2/p) \le \frac{2}{p} + \frac{2}{p(p-2)}$, from the Taylor series as in [7, (4.1)]. By (H3), the lower bound taken at $w^-$ and the upper bound at $z$,

$$2 \sum_{w \le p < z} \frac{1}{p} \le 2 \log \frac{\log z}{\log w} + \frac{1}{\log^2 z} + \frac{1}{\log^2 w} \le 2 \log \frac{\log z}{\log w} + \frac{2}{\log^2 Z},$$

and $\sum_{p \ge Z} \frac{2}{p(p-2)} \le T$. (ii) $w < Z$. Split the product at $Z$. The part over $w \le p < Z$ is at most $K_{\mathrm{ex}}(Z)(\log Z/\log w)^2$, and the part over $Z \le p < z$ is case (i) with $w = Z$ ($Z$ is even, hence not prime). Multiply. □

This is Cases 4 and 5 of [7, Lemma 4.1] with the cut at $Z$ instead of 293, which is what makes $K_{\mathrm{an}}$ small.

## 5. Proof of the main theorem

Throughout, $Q = \{3, 5\}$, $C_Q = 11$, $\ell_2 = \log 2$, $\ell_p = -\log(1-2/p)$ for odd $p$, and for $x > 2$

$$L(x) = \sum_{\substack{p < x \\ p \in \mathcal{P}}} \ell_p, \qquad E_\delta(x) = \sum_{\substack{p < x \\ p \in \mathcal{P}}} \log\bigl(1 + 8p^{-\delta}\bigr).$$

Thus $V(z) \ge e^{-L(z)}$ for every $N$.

### *5.1. A margin on an interval*

**Lemma 5.1.** *Assume Hypothesis 4.1. Let $I = (u, v]$ with $u \ge u_0$, let $K_I$ be as in Lemma 4.10, $k_I = 2 + \log K_I$, and let $s \ge 2k_I + 3$ with $F(s, k_I) > 0$ and $0 < \delta < 1$. Put $X_{\mathrm{lo}}(u) = (\sqrt{8}\,u^{16} - 1)/5 - 3$ and*

$$m_I(s, \delta) = \log X_{\mathrm{lo}}(u) - L(v) + \log F(s, k_I) - \log(2C_Q) - s\delta \log v - E_\delta(v).$$

*If $m_I(s, \delta) > 0$, then $S(\mathcal{A}, z) > 0$ for every odd $N$ with $z = (N^2/8)^{1/32} \in I$.*

*Proof.* Let $z \in I$ and $D = z^s$. By Lemma 4.10 and Theorem 2.2, $S(\mathcal{A}, z) \ge XV(z)F(s, k_I) - 2R_4(\mathcal{A}, D)$. Each factor is bounded at the appropriate end of $I$:

- $X \ge X_{\mathrm{lo}}(u)$ by Lemma 4.8, since $N = \sqrt{8}z^{16} > \sqrt{8}u^{16}$;
- $V(z) \ge e^{-L(v)}$, since $V$ decreases in $z$, the primes below $z$ are among those below $v$, and $g \le g^*$;
- $2R_4(\mathcal{A}, z^s) \le 2C_Q v^{s\delta} e^{E_\delta(v)}$ by (2.4), since $D^\delta = z^{s\delta} \le v^{s\delta}$ and the product increases in $z$.

Hence $\log\bigl(XV(z)F(s, k_I)/2R_4(\mathcal{A}, D)\bigr) \ge m_I(s, \delta) > 0$. □

### *5.2. The finite range $u_0 < z \le 10^8$*

**Proposition 5.2** (certified computation)**.** *There is a partition of $(u_0, 10^8]$ into 117 415 contiguous cells $I = (u, v]$, with parameters $(s_I, \delta_I)$ and constants $K_I$ as in Lemma 4.10, such that $s_I \ge 2k_I + 3$, $F(s_I, k_I) > 0$ and*

$$m_I(s_I, \delta_I) \ge 1.118347$$

*for every cell. The minimum is attained on the cell* $(10342.3367,\ 10343.3709]$*, with $s = 14.30443$, $\delta = 107/128$ and $K = 1.002250$.*

Code and output are listed under Data availability.

*Cells.* Geometric cells of ratio $1 + 10^{-4}$ from $u_0$ to $10^6$; above $10^6$, cells $[b_j, b_{j+1})$ with integer endpoints and relative step $2 \cdot 10^{-4}$ on $[10^6, 10^7]$ and $5 \cdot 10^{-4}$ on $[10^7, 10^8]$. The code checks that consecutive cells share endpoints, that the first starts at $u_0$ and that the last ends exactly at $10^8$.

*Dimension constant.* For cells below $10^6$, $K_I$ is computed from the exact candidates of Lemma 4.10 ($z \to u^+$, $z \to q^+$, $w$ prime), one ball per prime. For block cells $z \in [b_j, b_{j+1})$ we use

$$\log K_I \le L_{\mathrm{hi}}(b_{j+1}) - 2\log\log b_j + \max_{w<b_{j+1}} H(w), \qquad H(w) = 2\log\log w - L(w),$$

and, inside a block $[A, B)$, $H(w) \le 2\log\log B - L_{\mathrm{lo}}(A)$. A brute-force comparison at 7 points confirms that the rigorous bound is at least the exact value, with relative excess at most $1.1 \cdot 10^{-4}$.

*Parameters.* $\delta \in \{j/128 : 26 \le j \le 127\}$ (102 exact dyadic values); $s$ is optimised in floating point (grid and golden-section search) and then used as an exact constant. The conditions $s \ge 2k_I + 3$ and $F(s, k_I) > 0$ are verified in ball arithmetic on every cell.

*Ball arithmetic.* All bounds are evaluated with python-flint 0.9.0 (Arb) at 128 bits. Conversions from balls to floating point move one ulp outwards and are checked by exact Arb comparison.

*Sums over primes.* For $p < 10^6$ each term is evaluated separately as a ball. For $10^6 \le p < 10^{10}$ the primes are grouped into blocks $[A, B)$ for which the number $n$ of primes and $S_1 = \sum p$ are computed exactly (64-bit integers). For $f$ convex on $[A, B]$,

$$n\, f(S_1/n) \le \sum_{p\in[A,B)} f(p) \le n f(A) + \frac{(f(B) - f(A))(S_1 - nA)}{B - A}$$

(Jensen and the chord). The two functions used are convex: $\ell(x) = \sum_{j\ge1}(2/x)^j/j$ is a sum of convex functions for $x > 2$, and $e_\delta(x) = \log(1 + 8x^{-\delta})$ satisfies

$$e''_\delta(x) = \frac{8\delta x^{-\delta-2}\left[(\delta+1) + 8x^{-\delta}\right]}{(1 + 8x^{-\delta})^2} > 0.$$

The resulting enclosures are narrow: for $Q = \{3\}$,

$$L(10^8) \in [5.604898186,\ 5.604898205],$$

and the excess in $E_\delta(10^8)$ ranges from $6 \cdot 10^{-8}$ ($\delta \approx 0.99$) to $5 \cdot 10^{-3}$ ($\delta \approx 0.20$).

*Primes.* The primes up to $10^{10}$ are generated by a segmented sieve of Eratosthenes (20 730 blocks above $10^6$). The values $\pi(10^k)$ for $k = 6, \dots, 10$ agree with the known values (e.g. $\pi(10^{10}) = 455\,052\,511$), and the counts of 60 random blocks, of the four decades, and the exact sums of 12 blocks agree with the independent program primesieve 12.17, with no mismatch.

*Floating point versus certified.* The floating-point margins differ from the certified ones by at most $1.8 \cdot 10^{-13}$ over all covers; the certification guards against programming errors, the monotonicity arguments and the junction at $10^8$, not against rounding. Table 1 lists the minimal certified margin in each decade.

Table 1: Minimal certified margin $m_I$ per decade, $Q = \{3, 5\}$, per-cell $K$.

| range of $z$ | min. certified margin |
|---|---|
| $(39.9,\ 10^2]$ | +15.369 |
| $(10^2,\ 10^3]$ | +2.591 |
| $(10^3,\ 10^4]$ | +1.124 |
| $(10^4,\ 10^5]$ | **+1.118** |
| $(10^5,\ 10^6]$ | +1.829 |
| $(10^6,\ 10^7]$ | +3.414 |
| $(10^7,\ 10^8]$ | +5.510 |

The bottleneck is always $z \sim 1.0$–$1.6 \cdot 10^4$, i.e. odd $N \sim 10^{64}$–$10^{67}$. Figure 1 plots the certified margin of both chains over the whole finite range, and Figure 2 the per-cell constants $K_I$ that make it positive.

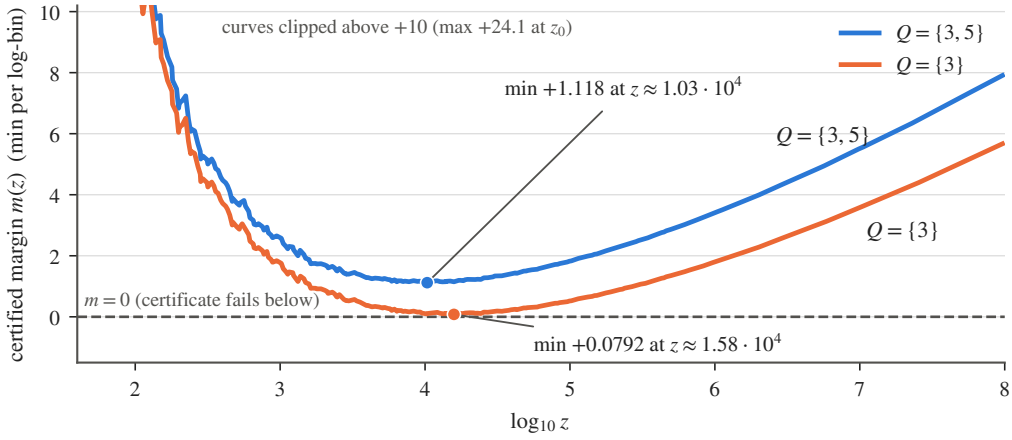



Figure 1: Certified margin $m_I$ on the finite range, per-cell $K$ (minimum over the cells in each of 400 logarithmic bins; ball-arithmetic lower bounds). Blue: $Q = \{3, 5\}$, 117 415 cells up to $10^8$; orange: $Q = \{3\}$, same range (122 023 cells up to $10^{10}$ in total). Both stay above 0; the dots are the minima of Table 3 (rows I). Nothing in this figure is floating-point only.

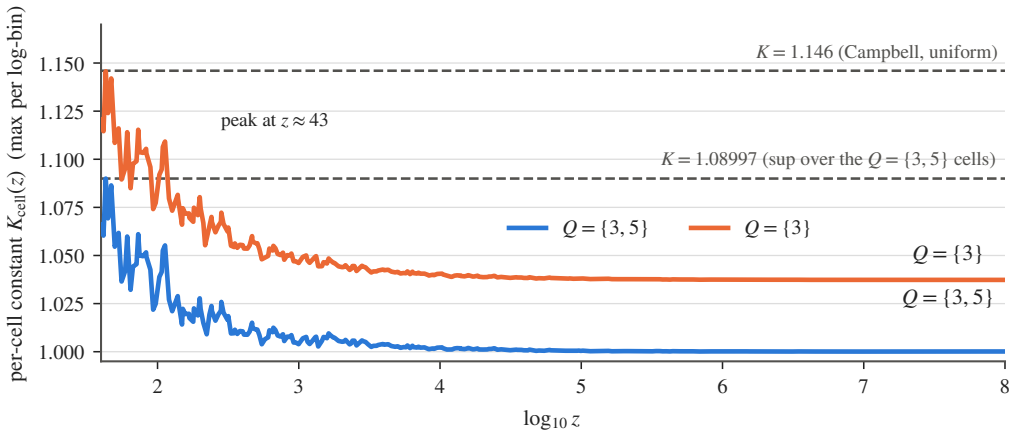



Figure 2: The certified upper bound $K_I$ for the dimension constant in each cell (maximum per bin). The per-cell values fall below 1.01 ($Q = \{3, 5\}$) and 1.04 ($Q = \{3\}$) after $z \approx 10^3$, while a uniform constant must cover the peak near $z \approx 43$: Campbell's 1.146 (dashed) and the supremum 1.08997 for $Q = \{3, 5\}$ both make the finite range fail (Section 6.2).

*Checks by separate implementations.* A separate floating-point implementation, not importing the main code (147 347 geometric cells with $\varepsilon = 10^{-4}$, up to exactly $10^8$), finds the minimum +1.1188 on (10342.337, 10343.371] with no failing cell. A per-prime evaluation without Jensen bounds (own sieve, mpmath) agrees with the certificate within $10^{-8}$ on 7 cells, and on the block cell (31 305 688, 31 321 341] the exact sum gives a larger margin than the certificate (6.683914226 > 6.683914135), as it must. The identity of Lemma 4.5 was checked exactly (rational arithmetic) for 859 odd $N$ and all $d \mid 2 \cdot 7 \cdot 11 \cdot 13 \cdot 17$, and Lemma 3.6 by brute force for odd $N \le 2000$, with no violation; these checks cover only small odd $N$.

*5.3. The analytic range $z \ge 10^8$*

Let $Z$ be an even integer with $Z \ge 286$, $t = \log z \ge \log Z$, and $K_{\mathrm{an}} = K_{\mathrm{an}}(Z)$ from Lemma 4.11.

**Lemma 5.3.** *For $z \ge Z$, $V(z) \ge c_V/\log^2 z$ with*

$$c_V = e^{-L(Z)} \log^2 Z \, e^{-2/\log^2 Z - T}.$$

*Proof.* $V(z) \ge e^{-L(Z)} \prod_{Z \le p < z}(1 - 2/p)$, and as in Lemma 4.11,

$$-\log \prod_{Z \le p < z} \Big(1 - \frac{2}{p}\Big) \le 2\log t - 2\log\log Z + \frac{2}{\log^2 Z} + T.$$

An upper bound $L_{\mathrm{hi}}$ for $L(Z)$ is used. □

**Lemma 5.4.** *Let $0 < \alpha < \log Z$, $\eta = \alpha/t$, $\delta = 1 - \eta$ and $D = z^s$. Then*

$$\log 2R_4(\mathcal{A}, D) \le \log(2C_Q) + st - \alpha s + 8\log t + 8\Big(B + \frac{1}{2t^2} - \sigma_Q\Big) + 8(e^\alpha - 1),$$

*where $\sigma_{\{3,5\}} = \frac{1}{3} + \frac{1}{5}$ (and $\sigma_{\{3\}} = \frac{1}{3}$).*

*Proof.* By (2.4) and $1 + x < e^x$, $R_4 \le C_Q D^\delta \exp\big(8\sum_{p<z,\, p\in\mathcal{P}} p^{-\delta}\big)$. For $p < z$ write $p^{-\delta} = p^{-1}e^x$ with $x = \eta \log p \in (0, \alpha)$. By convexity (the chord on $[0, \alpha]$, in place of $e^x < 1 + xe^x$ used in [7]), $e^x \le 1 + x\frac{e^\alpha - 1}{\alpha}$, so $p^{-\delta} \le \frac{1}{p} + (e^\alpha - 1)\frac{\log p}{pt}$. Summing with (H3) (upper bound) and (H4) gives $\sum_{p<z,\, p\in\mathcal{P}} p^{-\delta} \le \log t + B + \frac{1}{2t^2} - \sigma_Q + (e^\alpha - 1)$. Finally $D^\delta = e^{st - \alpha s}$. A term $-8(e^\alpha - 1)c_Q/t$ with $c_Q = \sum_{q\in Q} \log q/q$ has been dropped; it enters the margin with a positive sign, so dropping it is conservative. □

**Proposition 5.5.** *Assume Hypothesis 4.1. Let $Q = \{3, 5\}$, $Z = 10^8$, $k = 2 + \log K_{\mathrm{an}}$, and fix $s$ and $\alpha$ with $s \ge 2k + 3$ and $(16 - s)\log Z > 10$. Let $M(t)$ denote the logarithm of the ratio of $XV(z)F(s,k)$ to $2R_4(\mathcal{A}, z^s)$. Then $M(t) \ge M_{\mathrm{lo}}(\log Z)$ for all $t \ge \log Z$, where*

$$M_{\mathrm{lo}}(t) = 16t + \log\sqrt{8} - \log 5 + \log(1 - 16/N) + \log c_V - 10\log t + \log F(s,k) \\ - \log(2C_Q) - st + \alpha s - 8(B - \sigma_Q) - \frac{4}{t^2} - 8(e^\alpha - 1).$$

*With the constants of Table 2, $M_{\mathrm{lo}}(\log Z) \ge +0.654075$; hence $S(\mathcal{A}, z) > 0$ for every odd $N$ with $z \ge 10^8$.*

*Proof.* By Lemma 4.11, Theorem 2.2 applies at every $z \ge Z$ with $k = 2 + \log K_{\mathrm{an}}$. Write $N = \sqrt{8}e^{16t}$ and $X \ge (N - 16)/5$ (Lemma 4.8). Lemmas 5.3 and 5.4 give $M(t) \ge M_{\mathrm{lo}}(t)$; the factor $\log^{-2} z$ from $V$ contributes $-2\log t$ and $R_4$ contributes $-8\log t$. Then $M_{\mathrm{lo}}(t) = C_0 + (16 - s)t - 10\log t - 4/t^2 + \log(1 - 16/N)$ with a constant $C_0$, and

$$\frac{d}{dt}\Big[(16 - s)t - 10\log t - \frac{4}{t^2}\Big] = (16 - s) - \frac{10}{t} + \frac{8}{t^3} > 0$$

whenever $(16 - s)t > 10$, which holds for $t \ge \log Z$; the term $\log(1 - 16/N)$ is increasing in $t$. Hence $M_{\mathrm{lo}}$ is increasing on $[\log Z, \infty)$, with no grid. The value $M_{\mathrm{lo}}(\log Z)$ is evaluated in ball arithmetic. □

Table 2: Analytic range: constants and certified $M_{\mathrm{lo}}(\log Z)$. Rows marked "fixed $K$" replace $K_{\mathrm{an}}$ by the indicated constant. Negative values are not certified bounds; they say that the method fails with these parameters.

| $Q$ | $Z$ | $K_{\mathrm{ex}}(Z) \le$ | $K_{\mathrm{an}}$ | $c_V \ge$ | $s$; $\alpha$ | $M_{\mathrm{lo}}(\log Z)$ |
|---|---|---|---|---|---|---|
| $\{3,5\}$ | $10^8$ | 1.000061 | 1.005973 | 2.068826 | 14.24820; 0.57719 | **+0.654075** |
| $\{3\}$ | $10^8$ | 1.037250 | 1.043382 | 1.241296 | 14.39135; 0.58719 | −2.264 |
| $\{3\}$ | $10^9$ | 1.037245 | 1.042087 | 1.242838 | 14.38041; 0.58643 | **+0.371416** |
| $\{3\}$ | $10^{10}$ | 1.037243 | 1.041163 | 1.243941 | 14.37186; 0.58583 | +3.134105 |
| $\{3,5\}$, fixed $K = 1.08997$ | $10^8$ | | | | 14.56237; 0.59900 | −4.962 |
| $\{3\}$, fixed $K = 1.097$ | $10^9$ / $10^{10}$ | | | | | −3.680 / −1.458 |
| $\{3\}$, fixed $K = 1.146$ | $10^{10}$ | | | | | −5.280 |

For $Q = \{3,5\}$ and $Z = 10^8$ one has $(16 - s)\log Z = 32.269 > 10$, and $s \ge 2k + 3$ holds with room to spare. Figure 3 shows $M_{\mathrm{lo}}$ as a function of $z$.

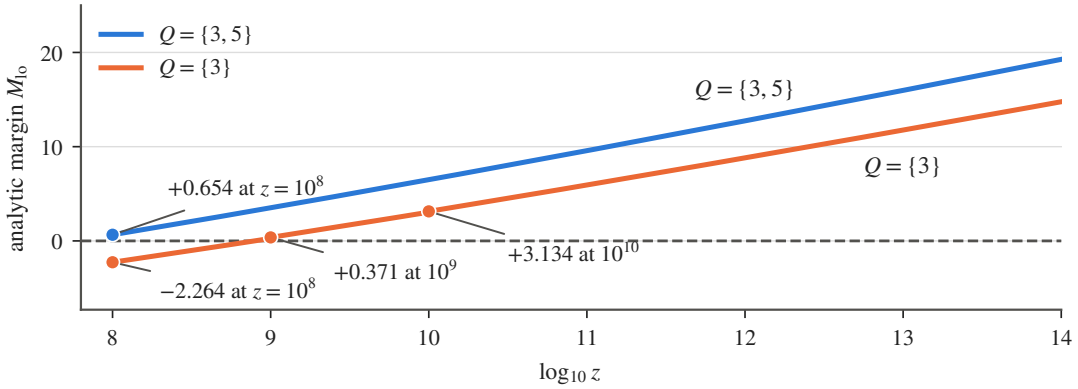



Figure 3: Analytic range: $M_{\mathrm{lo}}(t)$, $t = \log z$, with the constants and the $(s, \alpha)$ of Table 2 held fixed (ball-arithmetic lower endpoints; increasing in $t$ by Proposition 5.5). Blue: $Q = \{3,5\}$, constants of $Z = 10^8$. Orange: $Q = \{3\}$, upper envelope of the curves for $Z = 10^8, 10^9, 10^{10}$ (lower bounds valid for $z \ge Z$ may be combined by maximum). The dots are the entries of Table 2; the $Q = \{3\}$ curve is negative at $10^8$ and positive from $10^9$.

### *5.4. Completion of the proof*

Let $N \ge 2$. If $N \le N_0$, Corollary 3.2 applies. If $N > N_0$ is even, Proposition 3.4 applies. If $N > N_0$ is odd, then $z = (N^2/8)^{1/32} > u_0$. If $z \le 10^8$, $z$ lies in a cell of Proposition 5.2 and Lemma 5.1 gives $S(\mathcal{A}, z) > 0$; if $z \ge 10^8$, Proposition 5.5 gives $S(\mathcal{A}, z) > 0$. In both cases Lemma 3.7 yields $a, b$ with $N = a + b$ and $\Omega(ab) \le 32$. □

## 6. Remarks

### *6.1. A second chain with $Q = \{3\}$*

The theorem also follows from a chain that uses only the constants of [7, §3] ($C_Q = 19/6$) and not Section 4.2: the finite range up to $10^{10}$ with minimal margin +0.079236, and the analytic range from $Z = 10^9$ with +0.371416 (Tables 2 and 3); the two ranges overlap. The theorem holds if either chain is correct; the main chain of this paper is the one with $Q = \{3,5\}$ of Section 5.

Table 3: Minimal certified margins of the finite range, log(main/error). I: per-cell $K$; E: suffix-maximum envelope of the per-cell $K$; U: uniform $K$. Negative U values come from a grid search and mean that the method fails, not certified upper bounds.

| $K$ | $Q$ | up to | cells | failing | min. margin | cell | $s$ | $\delta$ | $K$ |
|---|---|---|---|---|---|---|---|---|---|
| I | $\{3,5\}$ | $10^8$ | 117 415 | 0 | **+1.118347** | (10342.3367, 10343.3709] | 14.30443 | 107/128 | 1.002250 |
| I | $\{3\}$ | $10^{10}$ | 122 023 | 0 | +0.079236 | (15822.3670, 15823.9492] | 14.44136 | 109/128 | 1.039801 |
| E | $\{3,5\}$ | $10^8$ | 117 415 | 0 | +1.005101 | same as I | | | 1.005973 |
| E | $\{3\}$ | $10^{10}$ | 122 023 | 0 | +0.036773 | same as I | | | 1.041163 |
| U ($K = 1.146$) | $\{3\}$ | $10^8$ | 117 415 | 85 342 | −3.405836 | $z \approx 1.18 \cdot 10^5$ | | | |
| U ($K = 1.089970$) | $\{3,5\}$ | $10^8$ | 117 415 | 71 182 | −1.671836 | $z \approx 4.29 \cdot 10^4$ | | | |

The envelope variant E, in which $K(z) = \sup_{z'' \geq z} K(z'')$, has the literal form "for every $z \geq z_a$" of the lemmas of [7]. It has the same logical status as the per-cell reading: both depend only on Hypothesis 4.1.

### 6.2. *A uniform constant does not suffice*

With Campbell's uniform $K = 1.146$ and optimised $s, \delta$, the finite range fails on 85 342 of 117 415 cells; with $Q = \{3,5\}$ and the uniform $K = 1.089970$ (the supremum over the range) it still fails (Table 3). So does the analytic range with fixed constants (Table 2). The per-cell constant, hence Hypothesis 4.1, is essential.

### 6.3. *Why $\Omega(ab) \leq 31$ does not follow*

Table 4 gives exploratory floating-point profiles (a grid of 60 points; these are not certified bounds) of the minimum of log(main/error) for the bound 31.

Table 4: Exploratory profiles for $\Omega(ab) \leq 31$ (floating point).

| case | relation | minimum | where |
|---|---|---|---|
| odd, $Q = \{3\}$ | $N = \sqrt{8}\, z^{15.5}$ | −5.312 | $z \approx 2.03 \cdot 10^5$; negative on $[534,\ 9.9 \cdot 10^7]$ |
| odd, $Q = \{3,5\}$ | same | −3.947 | $z \approx 5.9 \cdot 10^4$ |
| odd, $Q = \{3,5,7\}$ | same | −4.167 | pre-sieving more is worse |
| even, $Q = \{3\}$ | $N = 2z^{16}$ | −0.274 | negative on $[4659,\ 5.6 \cdot 10^4]$ |

The deficit of 4 to 5 in the logarithm for odd $N$ cannot be closed by constants; it seems to require a weighted sieve (Kuhn, Richert) or a better factor than $F(s,k)$, as suggested in [7, §6]. Figure 4 shows the profiles for $Q = \{3\}$ and $Q = \{3,5\}$.

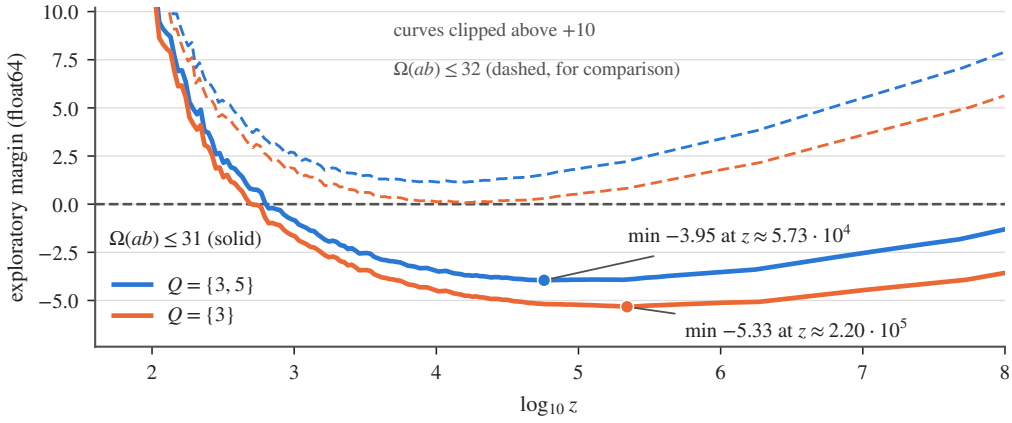



Figure 4: Why $\Omega(ab) \leq 31$ does not follow. Solid: exploratory *floating-point* margin for the bound 31 ($N = \sqrt{8}\, z^{15.5}$, float64, grid of 240 points, hence slightly below the 60-point minima of Table 4; not a certified bound), negative over most of the range with minima −3.95 ($Q = \{3,5\}$) and −5.33 ($Q = \{3\}$). Dashed: the same computation for the bound 32, which is positive (certified in Figure 1).

## Data and code availability

Code and certificate outputs are available at `https://github.com/giovanedms/goldbach-omega32` (Zenodo DOI to be assigned). The repository contains: `primes_blocks.py` (exact counts and sums of primes per block up to $10^{10}$), `check_blocks.py` (comparison with primesieve), `rlib.py` (ball arithmetic), `run_cover.py` (finite range), `analytic.py` (analytic range), `test_local.py` (brute force of the local lemmas), and their outputs. Environment: Python 3.14.4, python-flint 0.9.0, numpy 2.5.4, mpmath 1.3.0, primesieve 12.17; no GPU. The main finite cover (`run_cover.py 3,5 1e8 interval`) runs in about 40 s. Campbell's scripts [11] were re-run at commit `2755a58b6eb1...` and reproduce the versioned output byte for byte. SHA-256 of the main files:

```
7e8eb38ada4507a5599631cd2511db23d1d81d9f501a29d8c5da5e8163b1566c  out_cover_Q35_interval.txt
9d4dba8e248b89e284dbb3e57e16f2474b03b72bbb8725c30db6fa90e677259b  out_cover_Q3_interval.txt
b0d1624d1eb576a1ab9066db8abb06c54954598467d58a5e49382d23808bb0fa  out_analytic.txt
9e0bb1855b6a009c2bc5752961eae66a95c64904a220655429fbe01e64565f4d  rlib.py
9b3523413d3f70b9ff78a504c93ea44a1221022496badcfe035a0a2d1b40c10c  run_cover.py
e84fedd32399b8f56965a03f4f4cf7c992de248f03493b447d9b83edd4489eb3  analytic.py
4e901d77dfc171bce886f9924b9d8a0e59eb26b0b517a7ec53198cb7c552d86f  primes_blocks.py
```


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## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used [NAME OF TOOL / SERVICE] in order to [REASON]. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.


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## Declaration of competing interest

The author declares no competing interests.

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